Local laws, delocalization, and weak rigidity

A mathematical and Lean formalization guide

Paper-form theorems and independent kernel replay checked

Checked paper-form statements. Weak rigidity and all-vector delocalization use the literal structural event and its actual conditional uniform measure. A necessary correction to reference Definition 2.6 is stated below and proved consistent with the finite reference class. The main spectral declarations have no analytic certificate hypotheses. **Validation.** The independent integrated rebuild and trust-level-zero kernel replay have passed.

1 Objects, conditioning, and the scope of the reference class

Fix an integer $d \geq 3$. The probability space is the genuine uniform finite set of simple d-regular graphs on $\{0, \ldots, N-1\}$. Ensemble nonemptiness is explicit in the source theorems. Write $H = A/\sqrt{d-1}$ and $G(z) = (H - z)^{-1}$ for $\operatorname{Im} z > 0$. The paper's real-radius structural event is

$$\mathcal{F}_N(c, \omega) = \left\{ \max_v \operatorname{excess}(B_R(v)) \leq \omega, \quad \#\{v : B_R(v) \text{ is not a tree}\} \leq N^c \right\},$$

$$R = \frac{c \log N}{4 \log(d-1)}, \qquad 0 < c < 1.$$

Because graph distances and vertex counts are integers, the real-radius balls and the real count bound are proved exactly equivalent to the integer-floor definitions used by the local-law development. The final probability statement is under the actual uniform law on the structural subtype: for every fixed real $C > 0$, eventually in admissible N,

$$\mathbb{P}_{\mathrm{Unif}(\mathcal{F}_N)}(\text{failure}) = \frac{\mathbb{P}(\mathcal{F}_N \cap \{\text{failure}\})}{\mathbb{P}(\mathcal{F}_N)} \leq N^{-C}. \tag{1}$$

The event has mass at least $1/2$ eventually, so its nonemptiness and the denominator's positivity are proved. The final wrappers require no nonemptiness assumption on this structural subtype beyond eventual nonemptiness of the original regular-graph ensemble.

Necessary correction to Definition 2.6. Its printed estimate (2.10), $|G_{ii}(z)| \gtrsim 1$ for *all* $z \in \mathbb{C}_+$, is impossible: the genuine self-adjoint resolvent satisfies $|G_{ii}(\mathrm{i}Y)| \leq 1/Y$. The theorem `no_global_positive_diagonal_lower_bound` proves this obstruction. The global reference class therefore uses $|m_d(z)|/C_d \leq |G_{ii}(z)|$; it recovers constant lower and upper diagonal bounds on every fixed bounded spectral region. This correction is explicit, rather than a claim that the inconsistent printed definition was formalized unchanged.

The corrected `PaperReferenceThreshold` quantifies all connected, possibly infinite cores with finite deficit support and an actual finite feedback-edge set whose deletion leaves a forest. Their deficit sum plus feedback-edge count is at most ω. Its Green function is the genuine ℓ^2 resolvent of the literal infinite tree extension. Finite induced approximations, neighbor-closed embeddings, and strong resolvent convergence prove that this all-core class is equivalent to `FiniteReferenceThreshold`. The positive integer `paperOmega d` is independently defined as the maximum of this all-core class; the equality `paperOmega_eq_finiteOmega` is then proved. The final structural event uses this maximum, with no reference-class hypothesis left over.

```
SpectralConsequences.paperReferenceThreshold_iff_finite
SpectralConsequences.paperOmega_eq_finiteOmega
SpectralConsequences.allCore_reference_bounded_diagonal
SpectralConsequences.paperStructural_iff
SpectralConsequences.conditional_uniform_eq_subtype_uniform
```

The semicircle and Kesten–McKay density Stieltjes integrals are also identified with the actual scalar roots, rather than postulated.

2 Theorem 4.2 supplies the source local law

The existing `LocalLaw` project is the foundation. Its main declaration is

```
LocalLaw.source_theorem_4_2
LocalLaw.source_theorem_4_2_of_reference_threshold
```

It permits real a, b with $a \geq 12$ and $b \geq 25a$, and $0 < \beta \leq c/32$. The conclusion is uniform over the upper-half-plane domain $\operatorname{Im} z \geq (\log N)^b/N$, with the actual nominal model radius $r_{\mathrm{nom}} = \lfloor \beta \log N / \log(d-1) \rfloor$ and the article's actual error. All first- and second-stage graph resampling estimates used by this theorem are proved. No local-law, concentration, model-comparison, or resampling certificate is a hypothesis of its primary source declaration.

The proof uses a shorter internal model radius, losing $O(\log\log N)$ layers. It keeps the nominal error fixed, avoiding comparisons across the discontinuous error-inflation branches.

3 Theorem 1.4: all eigenvectors from Ward's identity

For a Hermitian matrix and an arbitrary eigenpair $Hv = \lambda v$, the proved deterministic criterion is

$$|v_i|^2 \leq \eta \operatorname{Im} G_{ii}(\lambda + \mathrm{i}\eta) \sum_j |v_j|^2, \qquad \eta > 0. \tag{2}$$

It includes the zero vector and does not assume a chosen eigenbasis or a simple eigenvalue. The source local law gives a degree-dependent uniform diagonal cap. With $a = 12$, $b = 300$, and $\eta = (\log N)^{300}/N$, this yields simultaneously for every eigenpair

$$|v_i|^2 \leq C_d \frac{(\log N)^{300}}{N} \sum_j |v_j|^2. \tag{3}$$

Consequently, every normalized eigenvector satisfies $\|v\|_\infty \leq \sqrt{C_d}(\log N)^{150}/\sqrt{N}$. Absorbing the fixed degree-dependent constant into one more logarithm gives the literal homogeneous conclusion

$$\|v\|_\infty \leq \frac{(\log N)^{151}}{\sqrt{N}} \|v\|_2.$$

The final predicate quantifies *all complex eigenvalues and all complex vectors*; Hermitian reality is proved for nonzero eigenvectors, and the zero vector is handled separately. No normalization, chosen eigenbasis, or simple-eigenvalue hypothesis is needed. The conclusion satisfies (1) on the literal structural subtype at `paperOmega d`. Its canonical wrapper takes only d, c, their numerical inequalities, and eventual ensemble nonemptiness; the auxiliary β is selected internally.

```
SpectralConsequences.eigenvector_coordinate_sq_le_green_im_mul_sum
SpectralConsequences.paperAllEigenvectors_iff_paperDelocalized
SpectralConsequences.full_paper_theorem_1_4
SpectralConsequences.full_paper_theorem_1_4_ratio
```

The deterministic criterion is in `EigenvectorWard.lean`. The source local-law application, constant absorption, unrestricted complex-eigenpair statement, and literal conditional uniform law are connected in the subsequent paper-form wrappers.

4 Counting, quantiles, and the distinction from confinement

The sorted eigenvalues are in decreasing order, with multiplicities retained. The constant vector supplies the distinguished index 0 eigenvalue $\tau_d = d/\sqrt{d-1}$. The other indices remain in the empirical measure even if their value also equals τ_d.

The actual Kesten–McKay density is normalized to mass one. Its compact CDF is continuous, monotone, and strictly increasing in the interior. The reference-mass estimate

$$F_d(y) - F_d(x) \geq c_*|y-x|^{3/2}, \qquad -2 \leq x \leq y \leq 2,$$

provides an inverse 2/3 modulus for its quantiles. Poisson smoothing converts the trace part of Theorem 4.2 into a uniform empirical-CDF bound; one checked source consequence is

$$\sup_E |F_N(E) - F_d(E)| \leq N^{-\beta/8}$$

with every polynomial conditional failure power.

A CDF estimate alone controls clamped eigenvalues, not outliers. The deterministic rigidity step explicitly also requires confinement. In a general form, CDF error δ and confinement error e imply

$$|\lambda_i - \gamma_i| \leq e + (\delta/c_*)^{2/3}. \tag{4}$$

In zero-based Lean indexing, the classical locations satisfy $F_d(\gamma_i) = (N-1-i)/N$. These are the exact source index levels after translating the article's one-based indexing. For one-based $2 \leq i \leq N$, the exact identity $\int_{\gamma_i}^2 \rho_d(x)\,dx = i/N$ is proved, including $\gamma_N = -2$. The source confinement event excludes only index 0. It does not discard every occurrence of the regular eigenvalue or assume connectedness.

```
SpectralConsequences.source_counting_rate
SpectralConsequences.empiricalCDF_error_of_trace_sqrt
SpectralConsequences.SquareRootCDF.inverse_modulus
SpectralConsequences.SquareRootCDF.rigidity_of_cdf_and_confinement
SpectralConsequences.RegularGraph.correctedSourceTrace_eq_positive_sum
```

The generic inverse-modulus and confinement-to-rigidity lemmas are in `QuantileRigidity.lean`; the source counting wrapper is in `SourceCountingProbability.lean`.

5 Section 7: physical fluctuations and reversible moments

5.1 The observables and the trivial mode

Let $q = Q(G,z)$ be the physical averaged oriented cavity observable, let $m = m_{\mathrm{sc}}(z)$, and let m_d be the Kesten–McKay transform. The depth ℓ is selected by the actual nonresonance rule. The finite cavity and full-root references are $Y_\ell(q) = \mathrm{cavityIterate}(z,q,\ell+1)$ and $X_\ell(q)$. The two corrected physical residuals are

$$F_Q = q - Y_\ell(q) - \delta_Q, \qquad F_m = m_N - X_\ell(q) - \delta_m.$$

The corrections are explicit physical tree-weight multiples of $\pi_N(z) = N^{-1}(\tau_d - z)^{-1}$. Tree root-to-boundary columns, the exact number of boundary labels, and the active labels after deleting a root neighbor are all proved for the actual graph balls. Thus δ_Q and δ_m are identified with genuine boundary-weight sums, including their multiplicities and zero inactive branches.

Subtracting the one distinguished pole gives a positive spectral sum over all indices other than 0. This is needed to exclude extra trivial eigenvalues as well as ordinary outliers. The scalar linearization and the two corrections obey a proved global cancellation estimate, including near τ_d; a pole-separation assumption is not imposed.

```
SpectralConsequences.actual_depth_trivial_correction_cancellation
SpectralConsequences.eventually_source_pole_scale_quadratic
SpectralConsequences.RegularGraph.cavity_boundary_all_label_weights
SpectralConsequences.RegularGraph.full_boundary_offdiag_correction
```

In particular $\Psi(z)|\pi_N(z)| \le C_{d,B}\varepsilon(z)^2$ on a bounded spectral domain. Near τ_d, the edge distance is positive; away from τ_d, the fractional error floor controls the inverse sample size. This turns the q-to-physical correction change into a quadratic drift cost.

5.2 Sampling geometry and exits

The resampling probability space includes both the uniformly random boundary labeling and the genuine independent edge-sampling fiber. Its switching map is reversible. Large protected tree patches imply actual acceptance of every label and preserve the same-radius excess bound and the root tree event. These are deterministic statements about the literal resampled graph, not assumed good-gadget conclusions.

At the smaller work parameter, two separate graph-count auxiliaries are used. The larger radius count bounds the failure of protected sampling; the smaller count preserves the original work bad-center threshold after switching. Both auxiliary complements have every polynomial graph-probability tail. The conditional geometric exit probability is at most $N^{-9/32}$. The auxiliary exponents are not used as a replacement for the target structural bad-count exponent.

`SpectralConsequences.RegularGraph.eventually_source_geometric_switch_exit_concrete`
`SpectralConsequences.LabelledResamplingState.ProtectedPatches.same_radius_excess`
`SpectralConsequences.LabelledResamplingState.ProtectedPatches.same_radius_root_tree`

5.3 What enters the genuine moment recurrence

The high-moment argument uses the complex test $T_p(F) = F^{p-1}\overline{F}^{\,p}$, so that $FT_p(F) = |F|^{2p}$. Exact finite disintegration lifts actual conditional entry means to the labelled-ball kernel. The resulting source drift integral for either observable is bounded by

$$C(\log N)^{6a+7}\varepsilon^2\,\mathbb{E}[\mathbf{1}_\Omega|F|^{2p-1}] + CM^{2p-1}N^{-D}. \tag{5}$$

The second term is a proved auxiliary-event loss. Uniform switched local caps are derived from the old event for *every* sampling assignment, so exceptional samples do not require an unproved new-graph resolvent cap.

On both old and new source events, genuine Ward estimates bound the corrected global increments by

$$V = C_{d,B}(\log N)^{2a}\frac{\operatorname{Im} m_d + \varepsilon + \Psi}{N\eta}. \tag{6}$$

The same variance satisfies $0 \le V \le \varepsilon/(\log N)^{16} \le \varepsilon$. Actual local observable increments are bounded by a fixed logarithmic multiple of ε. The exit integral is controlled separately, using the polynomial geometric exit and all-power weak-event/auxiliary failures. The recurrence keeps the mixed $\varepsilon_{\mathrm{loc}}V$ scale; it does not replace it by a first-order local-law error. This proved Section 7 generalization retains fixed logarithmic losses, including $(\log N)^{6a+7}\varepsilon^2$. They are absorbed into N^e in the weak-rigidity argument. It is not a claim of the printed log-free sharp Proposition 7.5 estimate.

`SpectralConsequences.global_reversible_moment_recurrence`
`SpectralConsequences.RegularGraph.eventually_source_moment_drift_integral`
`SpectralConsequences.RegularGraph.eventually_source_ward_variance_bound`
`SpectralConsequences.RegularGraph.eventually_source_ward_variance_small`
`SpectralConsequences.source_actual_high_moment_bounds`

The actual source high-moment family, its physical-tail bridge, and the final rigidity wrapper are checked. The reversible adapter requires increment bounds only on the nonzero support of the actual sampling law; zero-weight auxiliary states do not introduce a spurious hypothesis. No

intermediate moment or residual-tail predicate remains as a hypothesis of the final main theorem. The independent integrated rebuild and trust-level-zero kernel replay have also passed.

6 The final weak-rigidity source statement

For the original conditioning exponent c, set

$$c_{\mathrm{aux}} = \min(c/2, 1/4), \qquad \beta = \frac{c_{\mathrm{aux}}}{256}, \qquad \alpha = \frac{\beta}{32} = \frac{\min(c/2, 1/4)}{8192} > 0.$$

The source reduction uses $a = 12$, $b = 300$. Confinement from the corrected physical trace, combined with the checked counting law and (4), gives the weak-rigidity conclusion

$$\max_{1 \le i < N} |\lambda_i - \gamma_i| \le N^{-\alpha}$$

with the all-polynomial conditional bound (1). This is a weak positive-power rate, not the optimal edge $N^{-2/3}$ scale. The original conditioning event and its original radius remain in the final denominator, although the proof works internally at smaller radii.

The final public declarations are

```
SpectralConsequences.full_paper_theorem_1_2
SpectralConsequences.full_paper_theorem_1_2_ratio
SpectralConsequences.paper_classical_location_tail_integral
```

They are in `FullPaperTheorems.lean` and have only the numerical hypotheses $d \ge 3$, $0 < c < 1$, followed by eventual ensemble nonemptiness. They explicitly supply $\alpha > 0$, retain multiplicities, and use the literal one-based range $2 \le i \le N$. The primary form uses uniform measure on `PaperGraph`; the ratio companion is exactly equal to it by the proved conditioning identity. Earlier declarations such as `canonical_source_rigidity_of_residual_tails` remain useful intermediate composition lemmas, but their residual-tail hypothesis is discharged in the final theorem.

7 How to check and navigate the Lean development

The root directories `LocalLaw/` and `SpectralConsequences/` contain the local-law foundation and the spectral developments.
Final statements: `FullPaperTheorems.lean`.
Corrected all-core maximum: `PaperReferenceThreshold.lean`.
Each library's `Main.lean` and `Audit.lean` register its sources and query principal theorem axioms.
Ward: `EigenvectorWard.lean`; counting: `CDFComparison.lean`;
Section 7: `SourceMomentObservables.lean`.

From the repository root, the full check is

```
lake exe cache get
lake env python3 scripts/check.py --kernel
```

With compiled dependencies installed, `./check.sh -kernel` runs the same checker.
See `docs/BUILDING.md` for dependency overrides and offline use. Under `docs/`, `public-statements.txt`, `proof-map.txt`, and `paper-match-audit.txt` give the exact types, proof map, and source comparison.

The pinned Lean toolchain is 4.19.0 with its compatible Mathlib cache. The claimed source declarations use only `propext`, `Classical.choice`, and `Quot.sound`. The checker replays safe project declarations into a Mathlib base environment at kernel trust level zero. Lean's official replay excludes unsafe/partial compiler auxiliaries; all theorem proof terms are included in the safe mathematical declarations checked. There are no custom axioms, `sorry` placeholders, or assumed source mean/drift/exit certificates in these completed components. The source files and their audited declarations, rather than an informal proof outline, determine the exact checked statements.